

K-NEAREST NEIGHBOURS (K-NN) CLASSIFICATION

Given a dataset $\mathcal{D}$ consisting of n data points $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x is the input vector and y is the corresponding output label, the K-NN algorithm assigns a label to a new data point x_{new} based on the majority label of its K nearest neighbours in the dataset.

1. For the new data point x_{new} , compute its distance to all points in $\mathcal{D}$. Common distance measures include Euclidean, Manhattan, or Hamming distance.
2. Identify the K data points in $\mathcal{D}$ that are closest to x_{new} .
3. Assign the label y_{new} to x_{new} based on the majority label among the K nearest neighbors.

DISTANCE IN KNN (EUCLIDEAN DISTANCE)



For two points in a plane with coordinates (x_1, y_1) and (x_2, y_2) , the Euclidean distance between them is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$